

## Scaling India's Technology Ecosystem



**For Prelims:** Artificial Intelligence , Quantum Computing , PARAM program , India Semiconductor Mission , ONDC , Research, Development and Innovation (RDI) Scheme

**For Mains:** Emerging Technologies and India's Innovation Ecosystem, India's Transition from Innovation to Global Manufacturing Leadership.

Source: TH

## Why in News?

As India launches ambitious national missions in **Artificial Intelligence (AI), Semiconductors, Quantum Computing, and Space Technologies** , a critical debate has emerged regarding its innovation ecosystem. While **India boasts a rich history of scientific invention and early technological vision**, it has historically struggled to bridge the "**Valley of Death**" - the critical transition from **indigenous prototypes to globally dominant, commercialized industries**.

### Summary

- India has developed strong capabilities in emerging technologies such as AI, semiconductors, quantum computing, space technology, and Digital Public Infrastructure, but must overcome bottlenecks like low R&D spending, weak industry-academia linkages, and limited deep-tech financing to achieve global scale.
- Bridging the 'Valley of Death' through patient capital, translational research, strategic public procurement, technology diplomacy, and stronger innovation ecosystems will be crucial for positioning India as a global technology leader and achieving Viksit Bharat 2047.

## What Bottlenecks Limit India's Ability to Scale Innovations Globally?

- **Stagnant R&D Expenditure:** India's **Gross Expenditure on R&D (GERD)** has stagnated at roughly **0.64% of GDP** , compared to the **global average of ~1.7% and advanced economies (US/China) spending over 2.5%**.
- **Skewed Funding Pattern:** In leading innovative nations, the **private sector drives over 70% of**

## R&D.

- Despite policy interventions such as tax incentives and the establishment of the **Anusandhan National Research Foundation (ANRF)** designed to crowd-in private investment India's R&D burden remains disproportionately skewed **toward the public sector**.
- Consequently, scientific innovations often remain siloed within government research institutions, failing to translate into commercial viability.
- **The "Missing Middle" in Manufacturing:** India's industrial landscape comprises massive conglomerates and tiny informal MSMEs, **with a glaring lack of mid-sized, deep-tech firms** capable of building supply chains for complex innovations.
- **Weak Industry-Academia Linkage:** Scientific excellence in India often stays within **academic and public institutions (like CSIR or DRDO labs)** rather than transitioning into large industrial ecosystems.
- **Inadequate Venture Capital for Deep Tech:** While India has a vibrant **VC ecosystem** for software and consumer internet platforms, **"patient capital" required for hardware, deep-tech, and capital goods remains scarce**.
- **Intellectual Property (IP) Deficit:** India often excels in service delivery and process engineering but lags in originating and owning critical, standard-essential patents (SEPs) in frontier technologies.

# Historical Trajectory of Tech Innovation in India

Historically, India has suffered from the **"Stopping Too Soon" syndrome** , where success was measured by achieving indigenous capability rather than global commercial scale.

- **Semiconductor Complex Limited (SCL):** Established in the 1970s, India recognized the importance of integrated circuits early.
  - However, due to **limited capital, sub-optimal manufacturing scale, inward-looking public sector policies, and a lack of consistent support** , it failed to build a **global ecosystem akin to Taiwan's TSMC (Taiwan Semiconductor Manufacturing Company)** (world's leading producer of advanced semiconductors).
- **Electronics Corporation of India Limited (ECIL):** Set up in 1967 to counter technology embargoes, ECIL built indigenous computers and control systems.
  - However, its focus remained restricted to strategic state requirements rather than globally competitive commercial products.
- **The Simputer (1998):** An indigenous handheld device that anticipated modern **smartphone features** .
  - It failed to scale because the broader ecosystem of venture capital, component supply chains, and software platforms was immature.
  - Contrastingly, Apple leveraged **similar concepts by building a robust ecosystem to achieve global dominance.**

## Models of Success

- **Pharmaceutical Sector:** By leveraging process innovation and strict export orientation, **India scaled to become the "Pharmacy of the World" and a leading global vaccine manufacturer.**
- **Digital Public Infrastructure (DPI):** Platforms like **Aadhaar and Unified Payments Interface (UPI)** were designed for population scale from day one. They prove that scale creates massive ecosystems, which in turn drive global leadership.
- **PARAM Supercomputers:** The **PARAM program** proved that sustained public investment with clear, commercial performance goals can build globally recognized indigenous capabilities.
- **Space Sector (ISRO):** Through frugal engineering and commercial foresight, the **Indian Space Research Organisation (ISRO)** transformed from a domestic developmental agency into a global commercial launch provider.
  - By reliably deploying hundreds of foreign satellites via its **PSLV** and **LVM3** fleets and achieving milestones like the **Mars Orbiter Mission** at a fraction of global costs, India proved it can scale highly complex, cost-effective technological innovations for the global market.

# What are India's Opportunities in Emerging Technologies to Achieve Global Scale?

| Sector                  | India's Existing Strengths                                | Opportunities to Achieve Global Scale                                        |
|-------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| Artificial Intelligence | Large AI talent pool, Digital Public Infrastructure, vast | Develop <b>Sovereign AI</b> through low-cost, energy-efficient AI models, AI |

|                                            |                                                                                                                                                                   |                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(AI)</b>                                | multilingual datasets                                                                                                                                             | infrastructure, and domain-specific AI solutions for healthcare, education, agriculture, and governance.                                                                        |
| <b>Quantum Computing</b>                   | Strong research ecosystem, <b>National Quantum Mission</b> , expertise in mathematics and physics                                                                 | Build affordable quantum infrastructure and develop applications in <b>drug discovery, cybersecurity, materials science, financial modelling, and climate modelling</b> .       |
| <b>Semiconductors</b>                      | Around <b>20% of the global VLSI (Very-Large-Scale Integration) design workforce</b> , <b>India Semiconductor Mission (ISM)</b> , expanding electronics ecosystem | Scale <b>semiconductor fabrication (fabs), ATMP (Assembly, Testing, Marking and Packaging), chip design</b> , and indigenous semiconductor IP through targeted incentives.      |
| <b>Space Technologies</b>                  | Proven record of <b>frugal innovation</b> through <b>Chandrayaan , Mangalyaan</b> .                                                                               | Lead emerging areas such as <b>space-based data centres, orbital computing infrastructure, quantum communication, satellite manufacturing, and commercial launch services</b> . |
| <b>Digital Public Infrastructure (DPI)</b> | Successful platforms such as <b>Aadhaar, UPI, DigiLocker, ONDC, and CoWIN</b>                                                                                     | Export India's <b>Digital Public Infrastructure model</b> globally and develop interoperable digital public goods for developing countries.                                     |
| <b>Biotechnology &amp; Life Sciences</b>   | Global leadership in pharmaceuticals and vaccines, strong biotech industry                                                                                        | Expand into <b>genomics, precision medicine, synthetic biology, bio-manufacturing, and next-generation vaccine platforms</b> .                                                  |
| <b>Advanced Manufacturing</b>              | Strong engineering base and <b>Production Linked Incentive (PLI) schemes</b>                                                                                      | Promote <b>Industry 4.0, robotics, additive manufacturing (3D printing), advanced materials, and smart manufacturing</b> to enhance global competitiveness.                     |
| <b>Green &amp; Clean Technologies</b>      | Leadership in renewable energy deployment and green hydrogen initiatives                                                                                          | Become a global hub for <b>Green Hydrogen, battery storage, carbon capture, electric mobility, and climate technologies</b> .                                                   |
| <b>Cybersecurity &amp; Digital Trust</b>   | Growing digital economy and strong IT services ecosystem                                                                                                          | Develop indigenous <b>cybersecurity products, post-quantum</b>                                                                                                                  |

Click here to Read: [India's Emerging Technology Ecosystem](#)

## What Measures are Needed to Scale India's Emerging Technologies?

- **Bridge the "Valley of Death":** India has improved from **81st (2015)** to **38th (2025)** in the [Global Innovation Index 2025](#) , reflecting its emergence as a growing innovation hub. However, greater support is needed to scale research into commercially viable products.
  - To annex "Valley of Death" (the gap between research prototypes and commercial products) expand the [Research, Development and Innovation \(RDI\) Scheme](#) by establishing **Translational Research Centres** , as recommended in the [Economic Survey 2025-26](#) to provide shared prototyping and testing facilities for deep-tech startups.
- **Reform Academic R&D:** To shift from a **"Publish or Perish"** to a **"Patent & Produce"** ecosystem, India needs a robust statutory framework akin to the **US Bayh-Dole Act (1980)** , which allowed American universities to own, patent, and heavily monetize inventions created with federal funding, transforming them into commercial engines.
- **Provide Patient & Blended Finance:** Enable the [Anusandhan National Research Foundation](#) to deploy long-term, high-risk capital through industry-academia partnerships, mirroring Israel's Yozma initiative from the 1990s, which essentially created the Israeli tech-startup ecosystem.
- **Make Government the Anchor Customer:** Expand the [iDEX model](#) beyond defence and mandate a **5-10% public procurement quota** for indigenous deep-tech startups in sectors such as railways, healthcare, and smart cities, following the **NASA-SpaceX procurement model** .
- **Lead Global Technology Standards:** Financially support Indian firms to participate in [International Telecommunication Union \(ITU\)](#) , the [3rd Generation Partnership Project \(3GPP\)](#), and the [International Organization for Standardization \(ISO\)](#) , enabling them to secure **Standard Essential Patents (SEPs)** in emerging technologies such as **6G, AI, and Web 3.0** , transforming India from a **net importer to a net exporter of technology IP** .
- **Institutionalizing Brain Circulation:** Instead of trying to permanently stop emigration, India must leverage its diaspora. Schemes like [VAJRA \(Visiting Advanced Joint Research\)](#) and the [Ramanujan Fellowship](#) must be massively scaled to bring non-resident Indian scientists back as visiting professors, investors, and startup mentors, mirroring how **Taiwan utilized its Silicon Valley diaspora to build Hsinchu Science Park**.
- **Regulatory Sandboxes:** Implement **Regulatory Sandboxes** for emerging tech (drones, AI, crypto) so innovators can test products legally without the constant fear of sudden bans or retroactive taxation.
- **Tech-Alliances:** India must actively leverage diplomatic platforms like the [US-India iCET \(Initiative on Critical and Emerging Technology\)](#) and the [Quad](#) to secure technology transfers in defense, space, and quantum computing.
- **Mineral Diplomacy:** Expanding the mandate of the state-owned [KABIL \(Khanij Bidesh India Ltd\)](#) to aggressively acquire critical mineral assets in Africa, Australia, and South America to ensure an uninterrupted supply chain for battery and semiconductor manufacturing.

# Conclusion

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The countries that will lead the 21st century will not necessarily be those that invent first, but those that scale best. As India marches towards the goal of *Viksit Bharat 2047*, it must evolve its technological paradigm from mere self-reliance to aggressive global ambition, ensuring that Indian ingenuity is backed by robust ecosystems capable of conquering global markets.

## ***Drishti Mains Question:***

"India's technological history shows that being first matters little if you cannot scale." Discuss the structural bottlenecks in India's innovation ecosystem that prevent them from becoming global industries. Suggest measures to bridge the gap.

## Frequently Asked Questions (FAQs)

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### **1. What is the 'Valley of Death' in the innovation ecosystem?**

It refers to the stage between prototype development and commercialisation where deep-tech startups often fail due to inadequate funding, testing facilities, and market access.

### **2. What is the objective of the Research, Development and Innovation (RDI) Scheme?**

The RDI Scheme aims to promote indigenous innovation by supporting research, prototyping, commercialisation, and deep-tech entrepreneurship through long-term financing.

### **3. What is the significance of Standard Essential Patents (SEPs)?**

SEPs are patents essential for implementing global technology standards, enabling innovators to earn royalties and strengthen technological competitiveness.

### **4. How does the India Semiconductor Mission (ISM) support India's technology ecosystem?**

ISM promotes semiconductor fabrication, chip design, ATMP facilities, and indigenous semiconductor manufacturing to enhance technology self-reliance and supply-chain resilience.

### **5. What role does the Anusandhan National Research Foundation (ANRF) play in India's innovation ecosystem?**

ANRF promotes industry-academia collaboration by providing long-term funding for research, innovation, and commercialisation of emerging technologies.

# UPSC Civil Services Examination, Previous Year Questions (PYQs)

## Prelims

**Q. With reference to “Software as a Service (SaaS)”, consider the following statements: (2022)**

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

**Which of the statements given above are correct?**

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

**Ans: (d)**

## Mains

**Q.** COVID-19 pandemic has caused unprecedented devastation worldwide. However, technological advancements are being availed readily to win over the crisis. Give an account of how technology was sought to aid management of the pandemic. **(2020)**

## Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994



**Source: TH**

## Why in News?

The intersection of a **rising rural cancer burden** and rapid technological advancements in **portable ultrasonography** has brought **India’s Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994**, into sharp focus.

- A recent tragic case of a **rural patient succumbing to advanced breast cancer due to a lack of localized diagnostic access** highlights the urgent need to re-evaluate the regulatory limits placed on portable ultrasound machines.

# What is the PCPNDT Act, 1994?

- **About** : The PCPNDT Act aims to arrest the **declining Child Sex Ratio** and **combat female foeticide**.
  - It was substantially amended in 2003 to expand its regulatory scope over **pre-conception sex selection technologies**, strictly prohibiting the communication or disclosure of a foetus's sex under any circumstances.
- **Key Provisions:**
  - **Mandatory Registration & Equipment Control:** No facility can purchase or **operate ultrasound machines** without registering with the State's Appropriate Authority.
    - Once installed, moving the device outside the registered facility is a serious offense carrying a minimum of three months non-bailable imprisonment.
  - **Stringent Documentation:** Clinics must maintain meticulous records, notably **Form F** , which includes a declaration from the pregnant woman stating she does not wish to know the foetus's sex.
  - **Advertisement Ban:** Prohibits advertisements related to pre-conception or pre-natal sex determination.
  - **Supervisory Bodies** : Establishes the Central Supervisory Board, State Supervisory Boards, and Appropriate Authorities for implementation and monitoring.
  - **Penalties:** Offenses are cognizable, non-bailable, and non-compoundable. Violations attract up to three years of imprisonment, fines, and suspension of the medical practitioner's registration.
- **Impact of the Legislation:** Spurred by judicial interventions like the landmark **CEHAT v. Union of India (2001)** Supreme Court judgment mandating strict enforcement, the Act has yielded notable but mixed demographic results.
  - **Demographic Improvements:** The **Sex Ratio at Birth (SRB) improved to 929 females per 1,000 males at the national level (NFHS-5)** , up from 919 (NFHS-4).
    - It successfully curtailed the overt commercialization of sex-determination clinics.
  - **Unintended Consequences:** In some regions, the inability to access sex-selective abortions led to **increased fertility rates (families having children until a male is born)** and severe resource dilution.
    - This occasionally resulted in **higher child mortality for firstborn girls due to decreased parental investment in health**.
  - **Underground Networks:** The stringent ban inadvertently fueled a black **market utilizing unregistered portable machines and informal networks to evade authorities**.

## The Need for Legislative Reform

- **Technological Obsolescence:** The Act treats all ultrasound devices uniformly. Modern high-frequency linear probes used for **community-based cancer screening (e.g., detecting superficial breast lumps)** are physically incapable of foetal sex determination.
  - Yet, their movement is strictly prohibited, stifling localized healthcare delivery.
- **Rise of Advanced Genomics:** The current framework focuses heavily on ultrasound equipment and is ill-equipped to regulate **Non-Invasive Prenatal Testing (NIPT)**, which allows sex determination through a simple decentralized maternal blood draw.
- **Decriminalizing Clerical Errors:** Under Section 23 of the Act, minor clerical errors in "Form F" are often equated with the actual crime of sex determination.



- Although upheld by the Supreme Court in the ***Federation of Obstetric and Gynecological Societies of India (FOGSI) v. Union of India (2019)*** judgment, medical bodies argue this creates a chilling effect, deterring honest radiologists from providing essential obstetric care.
- **Integration with Artificial Intelligence (AI):** The law must adapt to permit AI-enabled systems. These devices can be software-locked to generate diagnostic reports for specific diseases while technologically blocking the visualization of foetal sex.

## Conclusion

The PCPNDT Act must be amended to allow portable, AI-assisted ultrasounds for rural cancer screening. Because modern high-frequency probes can be technologically locked to prevent foetal sex determination, policymakers can safely expand life-saving diagnostics to underserved populations without compromising the ongoing fight against female foeticide.

## Frequently Asked Questions (FAQs)

### 1. What is the primary objective of the PCPNDT Act, 1994?

The Act aims to prevent **sex selection and female foeticide** by regulating prenatal diagnostic techniques and prohibiting disclosure of the foetus's sex.

### 2. Why is there a demand to amend the PCPNDT Act?

The Act treats all ultrasound devices uniformly, restricting portable and AI-enabled ultrasound technologies that can improve rural diagnostics without enabling foetal sex determination.

### 3. What is Non-Invasive Prenatal Testing (NIPT)?

NIPT is a blood-based prenatal test that can detect fetal genetic abnormalities and, if misused, determine foetal sex, posing new regulatory challenges beyond conventional ultrasonography.

### 4. What was the significance of the **CEHAT v. Union of India (2001)** judgment?

The Supreme Court directed the strict enforcement of the PCPNDT Act to curb illegal prenatal sex determination and improve monitoring of diagnostic centres.

### 5. Why are AI-enabled ultrasound systems considered important for rural healthcare?

They can assist in early diagnosis of diseases such as breast cancer, improve community-level screening, and can be designed to technologically prevent foetal sex determination.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/wk5kyKzZP\\_U](https://www.youtube.com/embed/wk5kyKzZP_U) ]

## UPSC Civil Services Examination, Previous Year Questions (PYQs)

### *Mains*

**Q.** How do you explain the statistics that show that the sex ratio in Tribes in India is more favourable to women than the sex ratio among Scheduled Castes? **(2015)**

# Tungabhadra River Dispute



Source: TH

In a landmark move reflecting **cooperative federalism**, the Union Government has proposed a **High-Level Committee** following a rare consensus among **Karnataka, Andhra Pradesh, and Telangana** sparked by their joint meeting to inaugurate **33 newly installed spillway gates** at the **Tungabhadra Dam** to amicably resolve the **decades-old Tungabhadra basin water-sharing dispute**.

- **River Origin & Course:** The **Tungabhadra River** is a major perennial **right-bank tributary of the Krishna River** and a vital lifeline for the **Deccan Plateau**. It is formed by the confluence of the **Tunga** and **Bhadra** rivers, which originate in the **Western Ghats** near **Varaha Parvatha** and **Gangamoola** in Karnataka, and meet at **Kudli** near Shivamogga.
  - Flowing for about 531 km, the Tungabhadra passes through Karnataka, forms parts of the **Karnataka-Andhra Pradesh** and **Andhra Pradesh-Telangana** borders, and joins the Krishna River at Sangameswaram in Andhra Pradesh, serving as a vital lifeline for southern India.
  - The combined waters then flow eastwards and enter the **Bay of Bengal at Hamsaladeevi**.
- **The Tungabhadra Reservoir Project:** Commissioned in the 1950s near **Hosapete in Karnataka's Vijayanagara district**, this inter-state dam is the agricultural lifeline for drought-prone districts, irrigating over 16.38 lakh acres across the three states.
- **Historical Adjudication:** Water sharing was originally governed by the **Krishna Water Disputes Tribunal (KWDT-I, 1969)** led by **Justice R.S. Bachawat**, which allocated waters in a **65:35 ratio** between Karnataka and undivided Andhra Pradesh, managed via the **Tungabhadra Board**.
- **Current Allocations:** Following the 2014 state bifurcation, **Telangana holds an allocated entitlement of 15.9 TMC** from the Tungabhadra system.
- **Siltation Crisis:** Decades of heavy siltation have drastically reduced the dam's original **134 TMC storage capacity**, triggering severe downstream water deficits.
  - Infrastructural decay in the **Rajolibanda Diversion Scheme (RDS)** restricts Telangana from utilizing its full share, allowing it to draw only 5–6 TMC.
  - The **RDS** is an **interstate irrigation project of Karnataka and Telangana**, built across the **Tungabhadra River**. It diverts water through a **143-km canal** to irrigate drought-prone areas in **Karnataka, Telangana, and Andhra Pradesh**.
- **Upper Bhadra Project Conflict:** Andhra Pradesh has filed an original suit in the Supreme Court in 2023 against **Karnataka's upstream Upper Bhadra lift irrigation scheme**, arguing it threatens downstream flows to critical reservoirs like **Srisailem**.
- **Constitutional Distribution:** Water management falls under the **State List (Entry 17)**, but the regulation and development of inter-state rivers fall under the **Union List (Entry 56)**.
- **Dispute Resolution Framework:** **Article 262** empowers Parliament to adjudicate inter-state river disputes and allows it to **bar the jurisdiction of the Supreme Court**—a power exercised through the **Inter-State River Water Disputes (ISRWD) Act, 1956**.
- **Landmark Precedent:** The **Supreme Court's Cauvery Judgment (2018)** established the legal doctrine that **"inter-State rivers are national assets"**, rejecting exclusive state ownership and

mandating **equitable apportionment** .



Read more: [Tungabhadra Dam](#)

## NCERT Introduces Emergency in Class 9 Textbook



Source: TH

The **National Council of Educational Research and Training (NCERT)** has introduced the 1975 Emergency in its **Class 9 textbook, *Understanding Society: India and Beyond*** .

- The Emergency was imposed on 25th June 1975 on the grounds of internal disturbance and remained in force until March 1977.

## National Emergency

- **About: National Emergency (NE)** is proclaimed by the **President** under **Article 352** when the **security of India or a part of it** is threatened by **War, External Aggression (external emergency), or Armed Rebellion (internal emergency)**.

- The **38 th Amendment Act, 1975** , empowered the **President** to issue **multiple proclamations** on grounds of **war** , **external aggression** , **internal disturbance** , or **imminent danger** thereof, regardless of whether a prior proclamation is already in effect.
- Originally, the Constitution used **‘internal disturbance’** as the **third ground** for proclaiming a **National Emergency** , but the **44 th Amendment Act, 1978** , replaced it with **‘armed rebellion’** to remove its **vagueness** and make it **more objective**.
- **Territorial Extent:** NE can extend to the **whole of the country or only a part of it** . **42 nd Amendment Act, 1976** enabled the President to limit the operation of NE to a **specific part of India**.
- **Parliamentary Approval:** As per the **44 th Amendment Act** , 1978, a **NE** must be approved by **both Houses within one month** by a **special majority** (originally two months).
  - If the **Lok Sabha is dissolved** at the time of declaration, the **Rajya Sabha’s approval** remains valid, but the **reconstituted Lok Sabha** must approve it within **30 days of its first sitting**.
- **Duration:** It continues for **6 months** , and can be **extended to an indefinite period** with **approval of Parliament** for **every 6 months** (44 th Amendment Act 1978).
- **Revocation:** It can be **revoked anytime by the president** without the requirement of approval by Parliament.
  - The **Lok Sabha** can pass a **resolution to disapprove the continuation** of a **National Emergency** . If **one-tenth of its total members** submit a **written notice** to the **Speaker** (if in session) or to the **President** (if not in session), a **special sitting** must be held within **14 days** . The resolution must be passed by a **simple majority**.
- **Judicial Review:** The **38 th Amendment Act, 1975**, made the **Emergency declaration immune to judicial review** . This was later **reversed by the 44 th Amendment Act, 1978** .
  - In the **Minerva Mills case, 1980** , the Supreme Court held that a **Proclamation of NE can be challenged** if it is **mala fide** , based on **irrelevant or extraneous facts** , or is **absurd or perverse**.

Read More: [50 years of National Emergency](#)

## Textiles Summit 2026



**Source: PIB**

Recently, the Ministry of Textiles concluded the two-day Textiles Summit 2026, focused on sustainability, exports, global market expansion and India’s USD 100 billion textiles export target by 2030.

- **About:** Organised by the **Ministry of Textiles**, the summit brought together State Governments, industry, academia and Export Promotion Councils to prepare a National Textile Export Roadmap, with

emphasis on implementation, product-market alignment, value addition, niche products, high-value segments, quality, innovation and sustainability.

- **Sustainability and Traceability:** The summit stressed compliance with sustainability and environmental standards, quality certifications, digital product passports, traceability in the textile value chain, closed-loop recycling, circularity and better textile waste management through collaboration between municipal bodies and States.
- **Export Competitiveness:** Industry was urged to leverage recently concluded Free Trade Agreements, diversify export markets and products, focus on man-made fibres, strengthen Brand India and align national policies with evolving international regulatory standards.
- **Districts as Export Hubs:** States and UTs were encouraged to participate in the revived Districts as Export Hubs initiative, with a focus on addressing district-level information asymmetry and strengthening local export readiness.
- **Significance:** The summit supports India's USD 100 billion textile export target by 2030 through sustainability, traceability, circularity, MSME empowerment, FTA use, market access, quality, innovation and Brand India.

## Key Facts About Textile Industry

- The Indian textiles and apparel industry is valued at around USD 179 billion, contributing **about 2% to India's GDP**.
- India is the **6th largest global exporter** of textiles and apparel, with around 4% share in world exports.
- The sector is **India's 2nd-largest employment generator** after agriculture.
- **100% Foreign Direct Investment** is allowed in the textile sector under the automatic route.



Read More: [India's Textile Industry](#)



# BRICS Space Agencies Meeting Concludes in Bengaluru



**Source: PIB**

Union Minister of Earth Sciences pitched the idea of a “**BRICS Space Economy**” at the [BRICS](#) Heads of Space Agencies meeting in Bengaluru, calling for collective action to make space a driver of innovation, investment, entrepreneurship and sustainable development.

- **About:** The meeting was hosted by the [Indian Space Research Organisation](#) under India’s BRICS Chairship 2026 and attended by space agency heads and senior officials from Brazil, China, Egypt, Ethiopia, Indonesia, Iran, Russia, South Africa and the United Arab Emirates.
- **Purpose:** The meeting aimed to review progress in **BRICS space cooperation and explore ways to use space technology** for shared prosperity, resilience and solutions to global challenges.
- **Focus Area:** Deliberations covered space sustainability, debris-free missions, strengthening the **BRICS Remote Sensing Satellite Constellation** , participation of new BRICS members and the proposed BRICS Space Council.
- **Future Collaboration:** BRICS countries discussed cooperation in **disaster management, Earth observation, capacity building, knowledge sharing and satellite data sharing** among member countries.
- **India’s Space Ecosystem:** India showcased its growing **NewSpace sector**, private space industries and startups, along with achievements such as [Chandrayaan-3](#) , [Aditya-L1](#) and progress on the [Gaganyaan mission](#) .
- **Significance:** India called for BRICS space cooperation to move from **coordination to co-creation through joint innovation** , co-development, responsible space behaviour, sustainable space operations and stronger international partnerships.

**Read more:** [BRICS STI Cooperation](#)